AMPLE RELATIVE ANTI-CANONICAL DIVISORS AND FAILURE OF SUBADJUNCTION FOR MINIMAL LC CENTERS IN POSITIVE CHARACTERISTIC

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ABSTRACT. In this note, we construct algebraic fiber spaces $f : X \to Y$ in characteristic 2 or 3 from a projective 3-fold X to a smooth projective curve Y such that X has only canonical singularities and admits a crepant resolution and $-K_{X/Y}$ is ample. We also construct algebraic fiber spaces $f : X \to Y$ from projective lc 3-folds to smooth projective curves such that $-K_{X/Y}$ is ample and the unique lc center of X is a section of f. These examples violate the subadjunction formula for minimal lc centers.

1. Introduction

It is known that the relative anti-canonical divisor $-K_{X/Y}$ of an algebraic fiber space $f : X \to Y$ with suitable singularity assumptions is not ample unless f is constant. First, Kollár–Miyaoka–Mori [6, Corollary 2.8] proved the case where f is a smooth morphism. Miyaoka [7, Theorem 2] proved the result when f is generically smooth. Araujo–Druel–Kovács [1, Theorem 3.1] extended these theorems to lc pairs (X, Δ) when Y is a curve.

We recall the following generalizations of these results. Let k be an algebraically closed field of characteristic $p \geq 0$. Let (X, Δ) be a projective pair and let $f : X \to Y$ be a separable surjective morphism to a smooth projective variety Y of positive dimension.

Theorem 1.1 ([3, Corollary 3.7])**.** *Assume that $p = 0$ and (X, Δ) is lc over the generic point of Y. Then $-K_{X/Y} - \Delta$ is not ample.*

Corollary 1.2 ([2, Theorem 5.4])**.** *Assume that $p = 0$ and (X, Δ) is klt over the generic point of Y. Then $-K_{X/Y} - \Delta$ cannot be both nef and big.*

Theorem 1.3 ([2, Corollary 4.10])**.** *Assume that $p > 0$ and $(X_{\overline{\eta}}, \Delta|_{X_{\overline{\eta}}})$ is F-pure, where $X_{\overline{\eta}}$ is the geometric generic fiber of f. Then $-K_{X/Y} - \Delta$ is not ample.*

Corollary 1.4 ([2, Corollary 4.10])**.** *Assume that $p > 0$ and $(X_{\overline{\eta}}, \Delta|_{X_{\overline{\eta}}})$ is strongly F-regular. Then $-K_{X/Y} - \Delta$ cannot be both nef and big.*

Note that, when $p > 0$, we need to suppose that the geometric generic fiber has mild F-singularities. The F-purity (resp. strong F-regularity) of the geometric generic fiber implies the F-purity (resp. strong F-regularity) of the generic fiber, but the converse does not hold in general.

In this note, we consider the following question:

Question 1.5. Can the assumptions on the singularities of the geometric generic fiber in Theorem 1.3 and Corollary 1.4 be replaced by the corresponding assumptions on the generic fiber?

We give a *negative* answer to this question:

Theorem 1.6. (1) (*Example 2.4, $p = 2, 3$*) *There exists an algebraic fiber space $f : X \to Y$ from a projective 3-fold X to a smooth projective curve Y such that X has only canonical singularities and admits a crepant resolution and $-K_{X/Y}$ is ample.*
(2) (*Example 2.3, $p > 0$*) *There exists a strongly F-regular pair (P, Δ) of dimension 2 and an algebraic fiber space $f : P \to Y$ such that $-K_{P/Y} - \Delta$ is ample.*

Here, an algebraic fiber space is a separable surjective morphism with connected fibers. Although X in Theorem 1.6(1) is not F-pure, a crepant resolution $\tilde{X} \to X$ yields an algebraic fiber space $\tilde{f} : \tilde{X} \to Y$ from a smooth projective 3-fold $\tilde{X}$ to a smooth projective curve Y such that $-K_{\tilde{X}/Y}$ is semi-ample and big (hence nef and big).

The constructions in Theorem 1.6 are based on Raynaud's construction of surfaces [9] and its adaptation to dimension 3. A slight modification of the latter construction yields lc 3-folds (without boundary) violating the subadjunction formula for minimal lc centers. Before stating the theorem, we recall the subadjunction formula:

Theorem 1.7 ([5, Theorem 1], [4, Theorem 1.2], $p = 0$)**.** *Let $\mathbb{K}$ be $\mathbb{Q}$ or $\mathbb{R}$. Let X be a normal projective variety and let D be an effective $\mathbb{K}$-divisor on X such that (X, D) is lc. Let W be a minimal lc center with respect to (X, D). Then*

$$(K_X + D)|_W \sim_{\mathbb{K}} K_W + D_W$$

for an effective $\mathbb{K}$-divisor D_W on W such that (W, D_W) is Kawamata log terminal. In particular, W has only rational singularities.

Theorem 1.8 (Example 2.5, $p = 2, 3$)**.** *There exists an algebraic fiber space $f : X \to Y$ from a projective lc 3-fold X to a smooth projective curve Y such that*
(1) *$-K_{X/Y}$ is ample and*
(2) *the unique lc center W is a section of f.*
In particular, X violates the subadjunction formula for minimal lc centers, although W is smooth.

Indeed, we have

$$K_X|_W \sim (K_{X/Y} + f^*K_Y)|_W \sim \underbrace{K_{X/Y}|_W}_{\text{anti-ample}} + K_W,$$

so there exists no effective $\mathbb{K}$-divisor Δ_W with $K_X|_W \sim_{\mathbb{K}} K_W + \Delta_W$.

Declaration of AI assistance. The author used OpenAI's ChatGPT during the development of the example presented in this paper. In particular, ChatGPT was used to explore possible constructions and to assist with preliminary mathematical calculations. All mathematical statements and proofs presented in this paper have

been independently verified by the author, who takes full responsibility for their correctness.

2. Algebraic fiber spaces with ample relative anti-canonical divisors

We first consider a case in which Question 1.5 has an affirmative answer.

Notation 2.1. Let k be an algebraically closed field of characteristic $p \geq 0$. Let (X, Δ) be a projective pair and let $f : X \to Y$ be an algebraic fiber space to a smooth projective variety Y of positive dimension.

Lemma 2.2 ($p \geq 0$)**.** *Assume that* $\dim X - \dim Y = 1$ *and* $\mathrm{Supp}(\Delta)$ *does not dominate* Y*. Then* $-K_{X/Y} - \Delta$ *cannot be both nef and big.*

Proof. Suppose that $-K_{X/Y} - \Delta$ is nef and big. Then $-K_{X_{\overline{\eta}}}$ is ample, so $X_{\overline{\eta}} \cong \mathbb{P}^1_{\overline{K(Y)}}$, and hence $(X_{\overline{\eta}}, \Delta|_{X_{\overline{\eta}}})$ is klt (resp. strongly F-regular) when $p = 0$ (resp. $p > 0$). Note that $\Delta|_{X_{\overline{\eta}}} = 0$. Therefore, the statement follows from Corollaries 1.2 and 1.4. □

By Lemma 2.2, when $\Delta = 0$, there are no algebraic fiber spaces such that $\dim X = 2$, $\dim Y = 1$, and $-K_{X/Y}$ is both nef and big. We therefore consider the following two cases:

- The case when $\dim X = 2$ and $\Delta > 0$.
- The case when $\dim X = 3$, $\dim Y = 1$ and $\Delta = 0$.

In both cases, we can construct algebraic fiber spaces with mild singularities such that $-K_{X/Y} - \Delta$ is ample.

From now on, we use Notation 2.1 and assume that $p > 0$. We take a Tango–Raynaud curve $C = Y$ as the base. Let $F : Y \to Y$ denote the absolute Frobenius morphism. Let e, D and $C = Y$ be as in [8, Example 1.3]. Take $\xi \in H^1(Y, -D) \setminus \{0\}$ so that $F^*\xi = 0$. Let

$$0 \to \mathcal{O}_Y(-D) \to \mathcal{E} \to \mathcal{O}_Y \to 0$$

be the extension corresponding to ξ. Then, by the choice of ξ, we find a subbundle $\mathcal{O}_Y \subset F^*\mathcal{E}$.

Example 2.3 ($p > 0$)**.** We construct a klt pair (P, Δ) of dimension 2 and a surjective morphism $f : P \to Y$ such that $-K_{P/Y} - \Delta$ is ample. Set $P := \mathbb{P}(\mathcal{E})$. Let $f : P \to Y$ be the projection. Put $T := c_1(\mathcal{O}_P(1))$. Let G' be the prime divisor on $\mathbb{P}(F^*\mathcal{E})$ corresponding to the subbundle $\mathcal{O}_Y \subset F^*\mathcal{E}$. Set $G := F^*_{P/Y}G'$, where $F_{P/Y}$ is the relative Frobenius morphism of f. Then, by [8, Proposition 1.7], we see that G is smooth.

One can easily check that $G \sim pT$. Choose $\varepsilon \in \mathbb{Q}$ with $\frac{1}{p} < \varepsilon < \frac{2}{p} \leq 1$. Set $\Delta := \varepsilon G$. Then (P, Δ) is klt. Furthermore,

$$\begin{aligned} -K_{P/Y} - \varepsilon G &\sim_{\mathbb{Q}} 2T + f^*D - \varepsilon G \\ &\sim_{\mathbb{Q}} 2T + f^*D - p\varepsilon T \\ &\sim_{\mathbb{Q}} (2 - p\varepsilon)\, T + f^*D. \end{aligned}$$

Since $\delta := 2 - p\varepsilon \in (0,1)$, we have

$$-K_{P/Y} - \varepsilon G \sim_{\mathbb{Q}} \delta T + f^*D = \delta(T + f^*D) + (1-\delta)f^*D.$$

Since $T + f^*D$ is f-ample and nef and D is ample, we see that $-K_{P/Y} - \Delta$ is ample. Note that the divisor $G|_{P_{\overline{\eta}}}$ is p times a reduced point, so we see that $(P_{\overline{\eta}}, \varepsilon G|_{P_{\overline{\eta}}})$ is not F-pure.

From here on, we assume that $p = 2$ or 3. Put $l := 5 - p$.

Example 2.4 ($p = 2, 3$)**.** We construct an algebraic fiber space $f : X \to Y$ such that X is a canonical projective 3-fold having a crepant resolution, Y is a smooth projective curve, and $-K_{X/Y}$ is ample.

Let $C = Y$, D, $\mathcal{E}$ be as in Example 2.3. Set $\mathcal{G} := \mathcal{E} \oplus \mathcal{O}_Y(-D)$ and $P := \mathbb{P}(\mathcal{G})$. Let $h : P \to Y$ be the projection. Let F_1 be the prime divisor on P corresponding to the direct summand $\mathcal{O}_Y(-D) \subset \mathcal{G}$. Let G' be the prime divisor on $\mathbb{P}(F^*\mathcal{G})$ corresponding to the subbundle $\mathcal{O}_Y \subset F^*\mathcal{G}$, and let $G := F^*_{P/Y}G'$. Unlike the corresponding divisors in the surface construction, F_1 and G intersect. Put $T := c_1(\mathcal{O}_P(1))$. We have

$$F_1 + G \sim (p+1)T + h^*D \sim l(rT + h^*D'),$$

where $r := \frac{p+1}{l}$ and $D' := \frac{1}{l}D$. Note that D' is integral. Thus we can take the cyclic cover $\pi : X \to P$ of degree l branched over $F_1 + G$. Set $f := h \circ \pi$. Then,

$$\begin{aligned}
-K_{X/Y} &\sim \pi^*(-K_{P/Y} - (l-1)(rT + h^*D')) \\
&\sim \pi^*(3T + 2h^*D - (l-1)(rT + h^*D')) \\
&\sim \pi^*\left((3 - (l-1)r)T + \frac{l+1}{l}h^*D\right) \\
&\sim \pi^*\left(T + h^*(D + D')\right).
\end{aligned}$$

Since $\mathcal{G}(D + D')$ is an ample vector bundle, we see that $-K_{X/Y}$ is ample.

Next, we verify that X has a crepant resolution. Let Σ be the section of $h : P \to Y$ corresponding to $\mathcal{G} \twoheadrightarrow \mathcal{O}_Y(-D)$ and put $\Sigma_X := (\pi^{-1}(\Sigma))_{\mathrm{red}}$. Then one can check that

$$\mathrm{Sing}(X)_{\mathrm{red}} = (\pi^{-1}(F_1 \cap G))_{\mathrm{red}} \sqcup \Sigma_X.$$

Since F_1 and G are smooth and meet transversely along $F_1 \cap G$, the local equation of X over their intersection is

$$u^l = vw,$$

Thus X is étale locally

$$(A_{l-1} \text{ surface singularities}) \times (\text{smooth curve})$$

over $F_1 \cap G$, which has a crepant resolution.

Near Σ_X, the local equation of X is

$$u^l = x^p + ty^p.$$

When $(p, l) = (2, 3)$, we obtain

$$u^3 = x^2 + ty^2.$$

The blow-up along the ideal (x, y, u) is smooth and the discrepancy is 0. Thus, when $(p, l) = (2, 3)$, the crepant resolution $\tilde{X} \to X$ can be obtained by the blow-up along $\mathrm{Sing}(X)_{\mathrm{red}}$.

When $(p, l) = (3, 2)$, the local equation of X around Σ_X is

$$u^2 = x^3 + ty^3.$$

The y-chart of the same blow-up as above is

$$w^2 = y(v^3 + t)$$

with discrepancy 0. Taking $s = v^3 + t$, we get $w^2 = ys$, which is again $A_1 \times$(smooth curve). Hence it has a crepant resolution. Unlike the case $(p, l) = (2, 3)$, this case requires two successive blow-ups near Σ_X.

Example 2.5 ($p = 2, 3$)**.** We construct an algebraic fiber space $f : X \to Y$ such that X is an lc 3-fold, Y is a smooth projective curve, and $-K_{X/Y}$ is ample. Let $C = Y$, D, $\mathcal{E}$, $\mathcal{G}$, $h : P \to Y$, T, F_1, G, r and D' be as in Example 2.4. Let F_2 be the prime divisor on P corresponding to the subbundle $\mathcal{O}_Y(-D) \subset \mathcal{E} \subset \mathcal{G}$ of $\mathcal{G}$. Since we have

$$F_2 + G \sim (p+1)T + h^*D \sim l(rT + h^*D')$$

we can take the cyclic cover $\pi : X \to P$ of degree l branched over $F_2 + G$. Set $f := h \circ \pi$. By the same argument as in Example 2.4, we see that $-K_{X/Y}$ is ample.

Next, we check that X is lc. Let Σ and Σ_X be as in Example 2.4. Then $\Sigma = (F_2 \cap G)_{\mathrm{red}}$ and $F_2 + G$ is smooth outside Σ, so we have

$$\mathrm{Sing}(X)_{\mathrm{red}} = \Sigma_X.$$

Let $\mu : \tilde{P} \to P$ be the blow-up of P along Σ, and let E be its exceptional divisor. Put $B := F_2 + G$ and let $\tilde{B}$ be the strict transform of B. Then

$$K_{\tilde{P}} = \mu^* K_P + E \quad \text{and} \quad \mu^* B = \tilde{B} + (p+1)E,$$

and so

$$K_{\tilde{P}} + \frac{l-1}{l}\tilde{B} = \mu^*\left(K_P + \frac{l-1}{l}B\right) + \left(1 - \frac{l-1}{l}(p+1)\right)E$$

Let $\tilde{X}$ be the normalization of $X \times_P \tilde{P}$. We use the following notation:

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\beta} & X \\ {\scriptstyle\tilde{\pi}}\downarrow & & \downarrow{\scriptstyle\pi} \\ \tilde{P} & \xrightarrow[\mu]{} & P. \end{array}$$

Let R be the reduced exceptional divisor of β. A local computation shows that

- $\tilde{\pi} : \tilde{X} \to \tilde{P}$ is a cyclic cover of degree l branched over $\tilde{B}$,
- $\tilde{\pi}^* E = R$, and
- $\tilde{\pi}|_R : R \to E$ is a cyclic cover of degree l branched over $\tilde{B}|_E$. In particular, R is smooth, since $\tilde{B}|_E$ is smooth.

Therefore, by pulling back the above equation, we obtain

$$K_{\tilde{X}} = \beta^* K_X + \left(\frac{p+1}{l} - p \right) R.$$

Since $\{p, l\} = \{2, 3\}$, we get

$$K_{\tilde{X}} + R = \beta^* K_X.$$

Since $\beta : \tilde{X} \to X$ is a log resolution, we see that X is lc and $\beta(R) = \Sigma_X$ is the unique lc center. By construction, Σ_X is a section of f, which completes the proof.

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